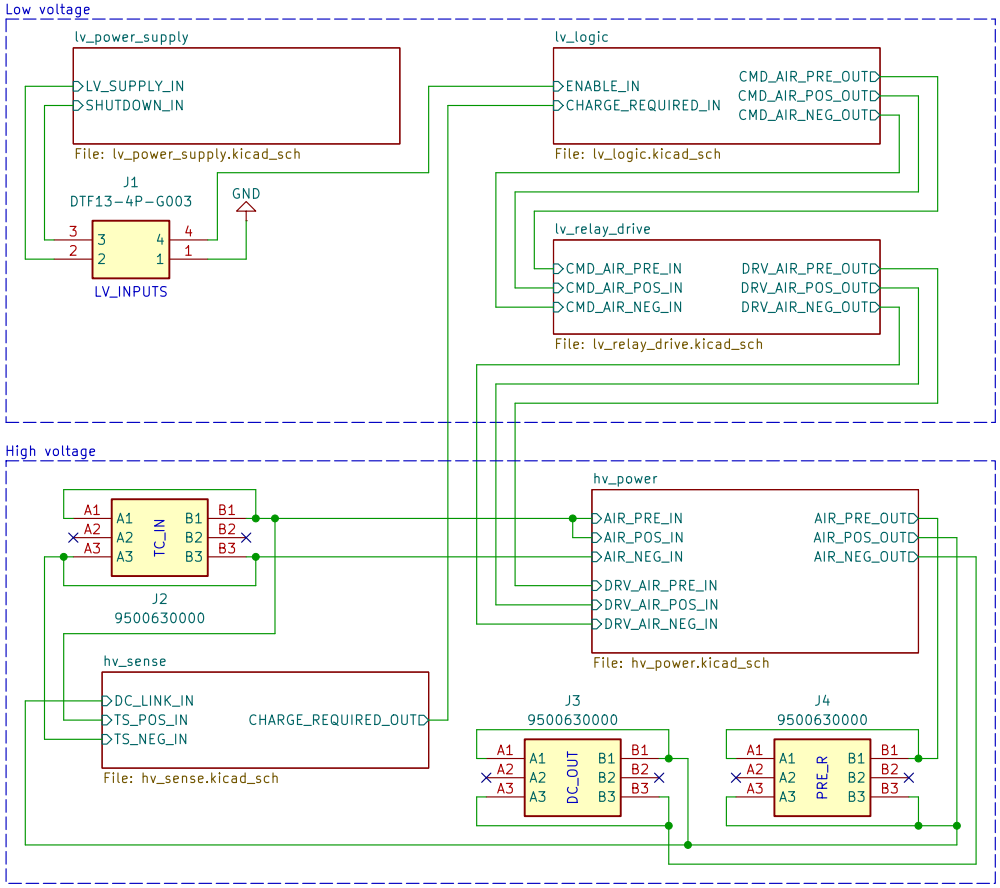


Sizing LV Connector (J1):
J1 = TE / Deutsch DTF13-4P-G003 (4-pin: LV_SUPPLY_IN, LV_SHDW_IN, ENABLE_IN, GND)
V_{LV_max} = 24V < max $V_{J1_nominal}$ = 250V DC
I_{LV_max} = I_{LOGIC_max} + I_{SHDW_peak} = 8mA + 470mA = 0.478A << max $I_{J1_nominal}$ = 13A (per contact)

Sizing HV Connectors (J2, J3, J4):
J2, J3, J4 = Weidmuller PCF 7.50/03/180 3.5SN OR BX (9500630000 – Flat-blade tab terminal)
V_{HV_max} = 600V DC <= max $V_{J_HV_nominal}$ = 630V / 690V (IEC)
$I_{HV_pre_peak}$ = 0.60A << max $I_{J_HV_nominal}$ = 24A
Center pin left unconnected -> hogher clearance

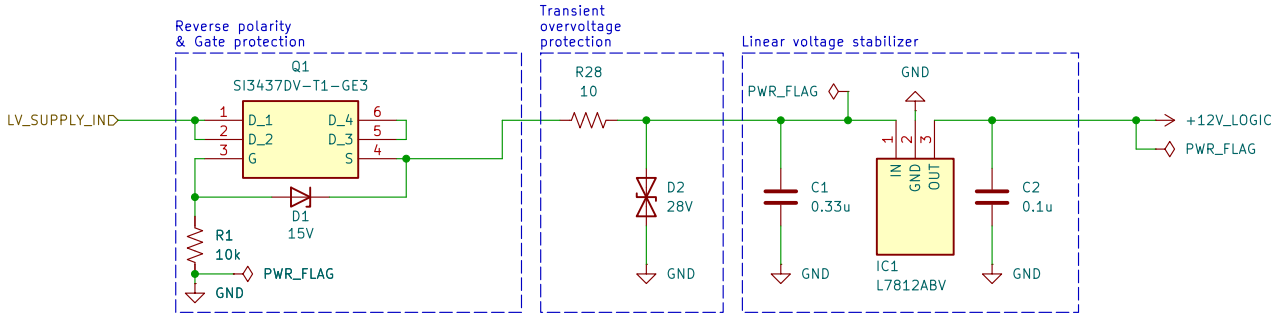
Hardware-only precharge controller design for the E-Agle Trento Racing Team electronics recruitment test.

By Bertagnolli Marco
University of Trento

Sheet: /
File: precharge-controller.kicad_sch

Title: Hardware precharge-controller

Size: A4	Date: 2026-11-01	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/6



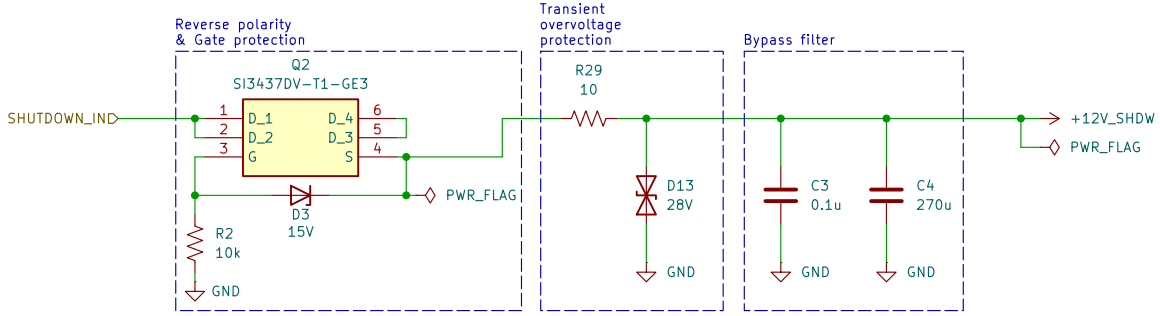
Input: V_{in} = 16V to 24V
Output: V_{out} = 12V

Max current draw:
$I_{logic_chips} = I_{IC2} + I_{IC3} + I_{IC5}$ = 5uA + 5uA + 5uA = 15uA (< 1mA)
$I_{PS1} = I_{DCDC_out}$ / eff = ((12V − V_{Q7}) / R9 + I_{IC4}) / 0.80 = ((12V − 1.2V) / 4.7kOhm + 0.8mA) / 0.80 = 3.88mA (~ 4.0mA)
$I_{max} = I_{IC1} + I_{logic_chips} + I_{PS1}$ = 4.3mA + 15uA + 3.88mA ~ 8mA

Sizing Reverse polarity & Gate protection:
Q1 = Si3437DV (max $|V_{DS}|$ = 30V > 24V, $R_{DS(on)}$ = 65mOhm, $P_D = R_{DS(on)} * I_{max}^2$ = 0.065Ohm * (8mA)² = 4.2uW)
R1 = 10kOhm (max $I_{R1} = (V_{in_max} - V_{D1})$ / R1 = (24V − 15V) / 10kOhm = 0.9mA, P_{R1} = 8.1mW)
D1 = 15V (max $V_{GSprotected}$ = 15V < max $V_{GSnominal}$ = 20V)

Sizing Transient overvoltage protection:
R28 = 10Ohm (V_{drop} = R28 * I_{max} = 10Ohm * 8mA = 80mV, $I_{clamp} = (V_{spike} - V_{D2clamp})$ / R28)
D2 = 28V (max $V_{L7812ABV_INprotected} = V_{D2clamp}$ = 28V < max $V_{L7812ABV_INnominal}$ = 35V)

Sizing Linear voltage stabilizer:
IC1 = L7812ABV ($P_{dis} = (V_{in_max} - V_{out}) * I_{logic_max} + V_{in_max} * I_{IC1}$ = (24V − 12V) * 4mA + 24V * 4.3mA = 151mW)
C1 = 0.33uF as L7812ABV's datasheet directions (multilayer ceramic)
C2 = 0.1uF as L7812ABV's datasheet directions (multilayer ceramic)



Input: V_{in} = 12V
Output: V_{out} = 12V

Max current draw:
I_{hold_coil} = 117mA at 6V (I_{peak_coil} = 12V / 51Ohm = 235mA at 12V)
$I_{max} = I_{U2} + I_{U3}$ = 117mA + 117mA = 234mA (max 2 relays energized simultaneously)

Sizing Reverse polarity & Gate protection:
Q2 = Si3437DV (max $|V_{DS}|$ = 30V > 12V, $R_{DS(on)}$ = 65mOhm, $P_D = R_{DS(on)} * I_{max}^2$ = 0.065Ohm * (234mA)² = 3.56mW)
R2 = 10kOhm (max $I_{R2} = V_{in}$ / R2 = 12V / 10kOhm = 1.2mA, P_{R2} = 14.4mW)
D3 = 15V (max $V_{GSprotected}$ = 15V < max $V_{GSnominal}$ = 20V)

Sizing Transient overvoltage protection:
R29 = 10Ohm (V_{drop} = R29 * I_{max} = 10Ohm * 234mA = 2.34V, $I_{clamp} = (V_{spike} - V_{D13clamp})$ / R29)
D13 = 28V (max $V_{SHDWprotected} = V_{D13clamp}$ = 28V < max $V_{DS_drivernominal}$ = 60V)

Sizing Bypass filters:
C3 = 0.1uF high-frequency EMI filter (multilayer ceramic)
C4 = 330uF ($C_{min} = (I_{max} * dt)$ / dV = (234mA * 0.5ms) / 0.5V = 234uF < C4 = 330uF)

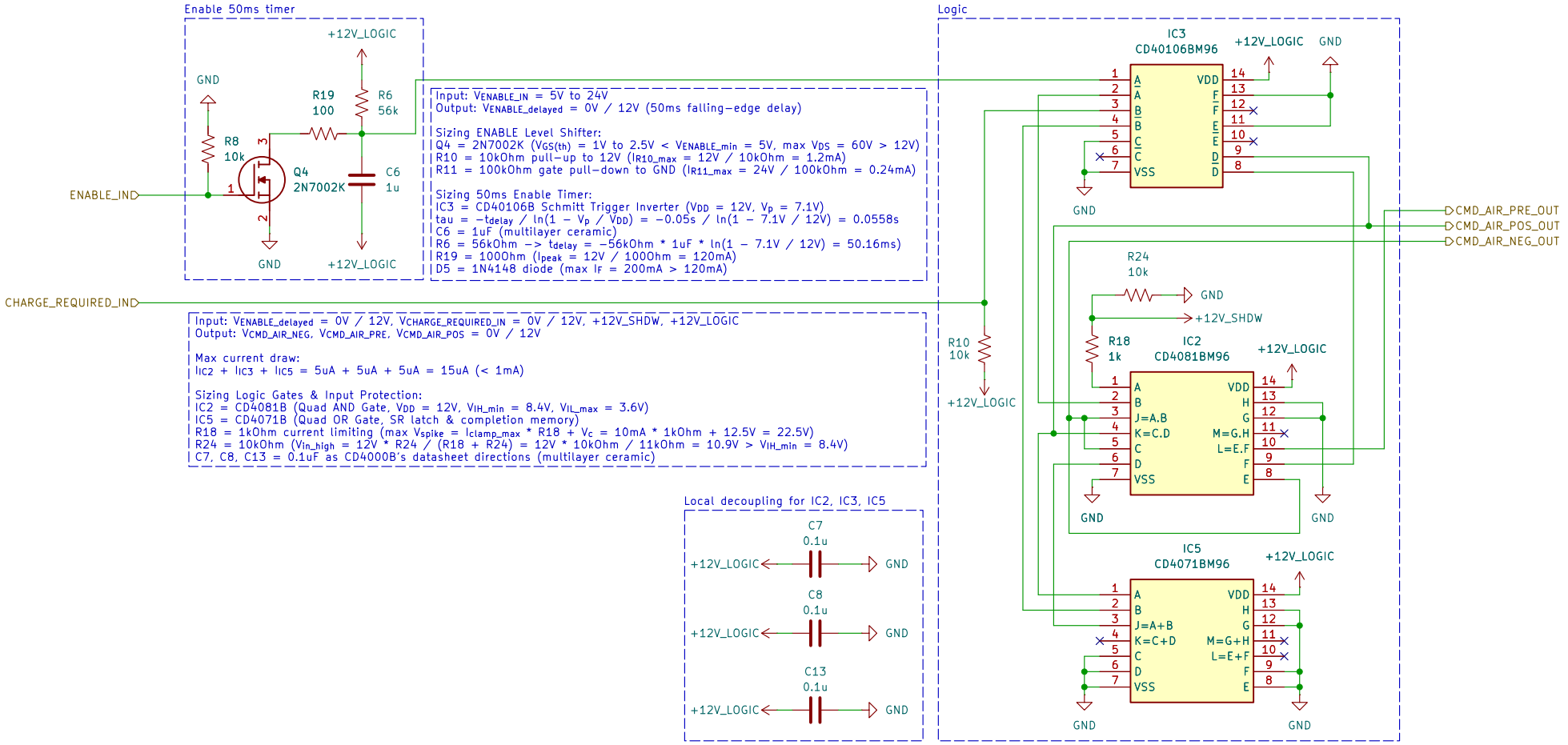
Hardware-only precharge controller design for the E-Agle Trento Racing Team electronics recruitment test.

By Bertagnolli Marco
University of Trento

Sheet: /lv_power_supply/
File: lv_power_supply.kicad_sch

Title: Hardware precharge-controller

Size: A4	Date:	**Rev: 1.0**
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Hardware-only precharge controller design for the E-Agle Trento Racing Team electronics recruitment test.

By Bertagnolli Marco

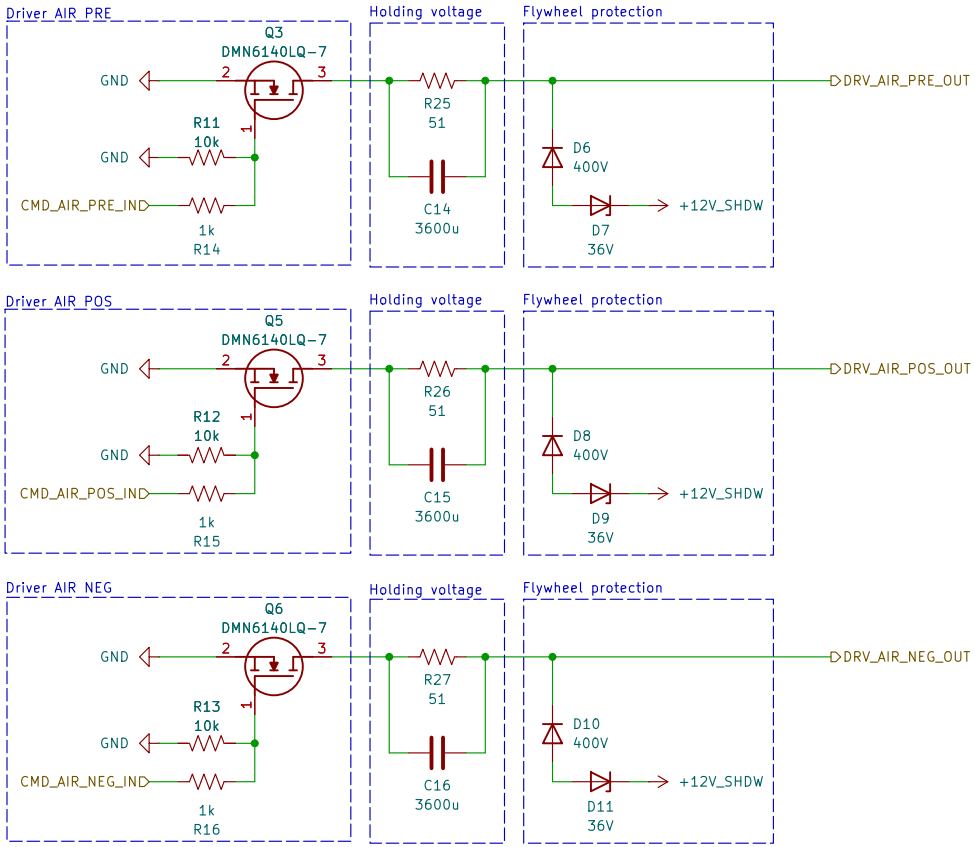
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Sheet: /lv_logic/

File: lv_logic.kicad_sch

Title: Hardware precharge-controller

Size: A4	Date:	**Rev: 1.0**
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Input: $V_{CMD_AIR_NEG}$, $V_{CMD_AIR_PRE}$, $V_{CMD_AIR_POS}$ = 0V / 12V
Output: Low-side drive for U2, U3, U4 coils (+12V_SHDW supply)

Max current draw (per channel):
$I_{coil_peak} = V_{SHDW} / R_{coil}$ = 12V / 510hm = 235mA
$I_{coil_hold} = V_{hold} / R_{coil}$ = 6V / 510hm = 117mA

Sizing Low-Side NMOS Driver:
Q3, Q5, Q6 = DMN6140L (max V_{DS} = 60V, $R_{DS(on)}$ = 50mOhm, $P_{D_hold} = R_{DS(on)} * I_{coil_hold}^2$ = 0.050hm * (117mA)^2 = 0.68mW)
R14, R15, R16 = 1kOhm gate resistor (I_{G_peak} = 12V / 1kOhm = 12mA)
R11, R12, R13 = 10kOhm gate pull-down to GND (prevents floating gate)

Sizing Coil Holding Voltage Network (Omron G9KB-1A12V, R_{coil} = 510hm):
R25, R26, R27 = 510hm ($V_{hold} = V_{SHDW} * R_{coil} / (R_{coil} + R_{hold})$ = 12V * 510hm / 1020hm = 6V, P_R = (117mA)^2 * 510hm = 0.70W < 2W)
$R_{eq} = (R_{coil} * R_{hold}) / (R_{coil} + R_{hold})$ = (510hm * 510hm) / (510hm + 510hm) = 25.50hm
C14, C15, C16 = 3300uF ($C_{calc} = -t_{pull} / (R_{eq} * \ln(1 - V_C / V_{max}))$ = −0.1s / (25.50hm * ln(1 − 4.2V / 6V)) = 3260uF -> 3300uF)

Sizing Freewheeling & Back-EMF Protection:
D6, D8, D10 = ES1G freewheeling diode (max I_F = 1A > I_{coil_peak} = 235mA)
D7, D9, D11 = 36V Zener clamp (max $V_{DS\,protected} = V_{SHDW} + V_Z + V_F$ = 12V + 36V + 1V = 49V < max $V_{DS\,nominal}$ = 60V)

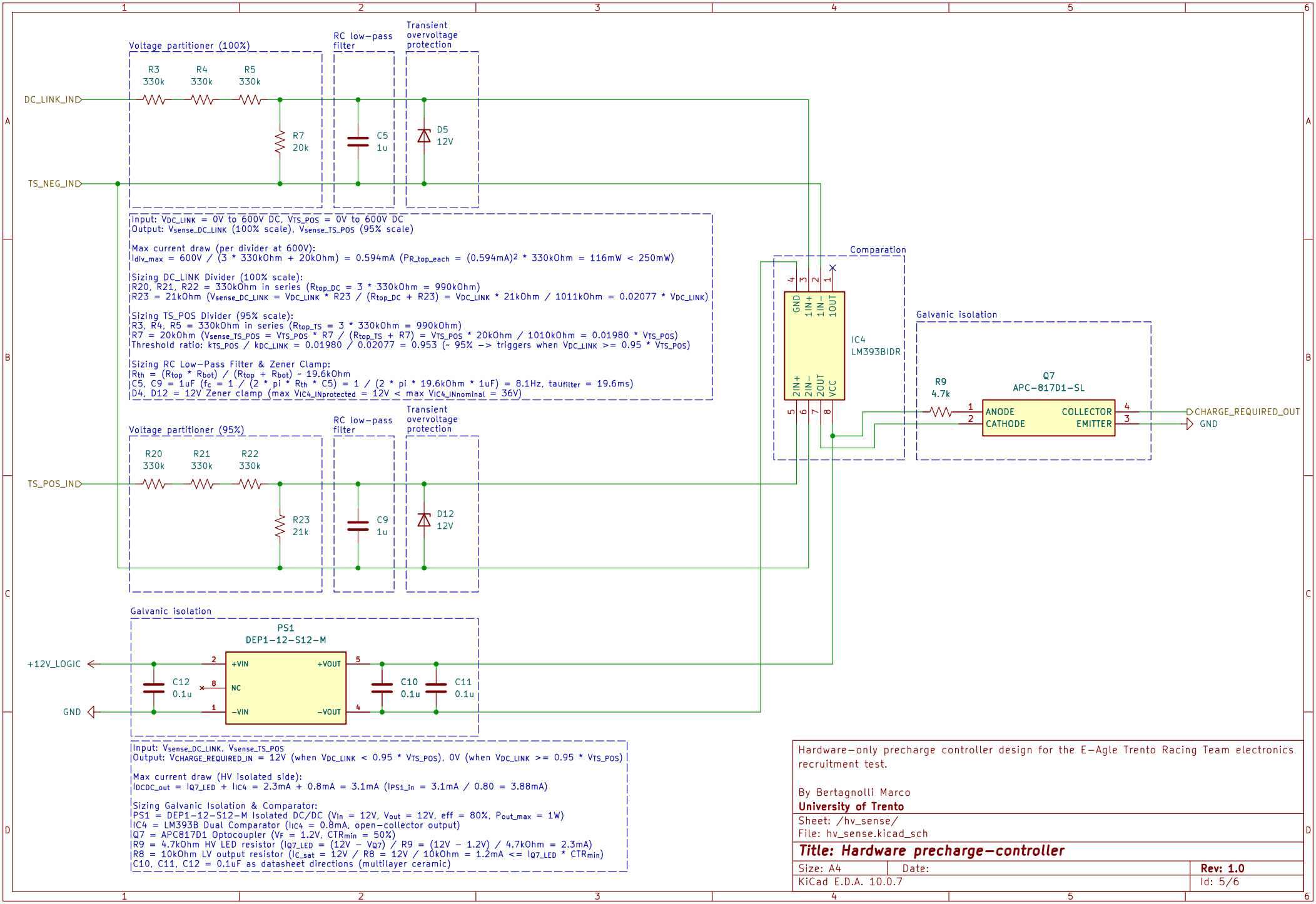
Hardware-only precharge controller design for the E-Agle Trento Racing Team electronics recruitment test.

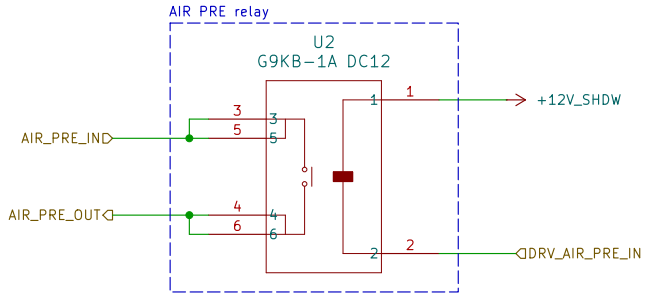
By Bertagnolli Marco
University of Trento

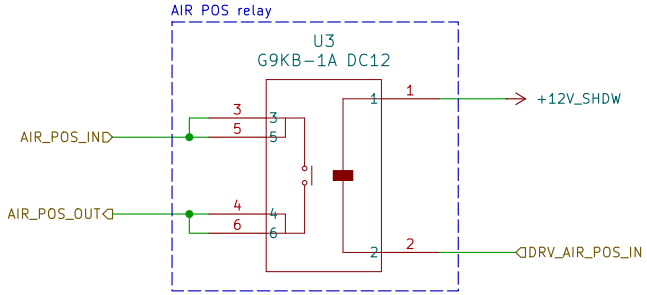
Sheet: /lv_relay_drive/
File: lv_relay_drive.kicad_sch

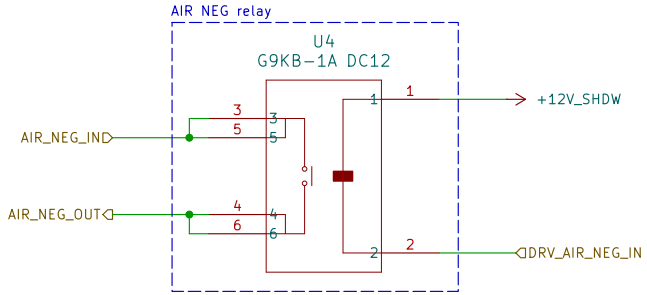
Title: Hardware precharge-controller

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Voltage partitioner (100%)
RC low-pass filter
Transient overvoltage protection
R3 330k
R4 330k
R5 330k
R7 20k
C5 1u
D5 12V
DC_LINK_IND
TS_NEG_IND
Input: VDC_LINK = 0V to 600V DC, VTS_POS = 0V to 600V DC
Output: Vsense_DC_LINK (100% scale), Vsense_TS_POS (95% scale)
Max current draw (per divider at 600V):
Idiv_max = 600V / (3 * 330kOhm + 20kOhm) = 0.594mA (PR_top_each = (0.594mA)2 * 330kOhm = 116mW < 250mW)
Sizing DC_LINK Divider (100% scale):
R20, R21, R22 = 330kOhm in series (Rtop_DC = 3 * 330kOhm = 990kOhm)
R23 = 21kOhm (Vsense_DC_LINK = VDC_LINK * R23 / (Rtop_DC + R23) = VDC_LINK * 21kOhm / 1011kOhm = 0.02077 * VDC_LINK)
Sizing TS_POS Divider (95% scale):
R3, R4, R5 = 330kOhm in series (Rtop_TS = 3 * 330kOhm = 990kOhm)
R7 = 20kOhm (Vsense_TS_POS = VTS_POS * R7 / (Rtop_TS + R7) = VTS_POS * 20kOhm / 1010kOhm = 0.01980 * VTS_POS)
Threshold ratio: kTS_POS / kDC_LINK = 0.01980 / 0.02077 = 0.953 (~ 95% -> triggers when VDC_LINK >= 0.95 * VTS_POS)
Sizing RC Low-Pass Filter & Zener Clamp:
Rth = (Rtop * Rbot) / (Rtop + Rbot) ~ 19.6kOhm
C5, C9 = 1uF (fc = 1 / (2 * pi * Rth * C5) = 1 / (2 * pi * 19.6kOhm * 1uF) = 8.1Hz, taufilter = 19.6ms)
D4, D12 = 12V Zener clamp (max VIC4_INprotected = 12V < max VIC4_INnominal = 36V)
Voltage partitioner (95%)
RC low-pass filter
Transient overvoltage protection
R20 330k
R21 330k
R22 330k
R23 21k
C9 1u
D12 12V
TS_POS_IND
Comparation
GND
1IN+
1IN-
1OUT
2IN+
2IN-
2OUT
VCC
IC4
LM393BIDR
Galvanic isolation
R9 4.7k
Q7
APC-817D1-SL
ANODE
CATHODE
COLLECTOR
EMITTER
CHARGE_REQUIRED_OUT
GND
Galvanic isolation
PS1
DEP1-12-S12-M
+VIN
NC
-VIN
+VOUT
-VOUT
+12V_LOGIC
GND
C12 0.1u
C10 0.1u
C11 0.1u
Input: Vsense_DC_LINK, Vsense_TS_POS
Output: VCHARGE_REQUIRED_IN = 12V (when VDC_LINK < 0.95 * VTS_POS), 0V (when VDC_LINK >= 0.95 * VTS_POS)
Max current draw (HV isolated side):
IDCDC_out = IQ7_LED + IIC4 = 2.3mA + 0.8mA = 3.1mA (IPS1_in = 3.1mA / 0.80 = 3.88mA)
Sizing Galvanic Isolation & Comparator:
PS1 = DEP1-12-S12-M Isolated DC/DC (Vin = 12V, Vout = 12V, eff = 80%, Pout_max = 1W)
IC4 = LM393B Dual Comparator (IIC4 = 0.8mA, open-collector output)
Q7 = APC817D1 Optocoupler (VF = 1.2V, CTRmin = 50%)
R9 = 4.7kOhm HV LED resistor (IQ7_LED = (12V - VQ7) / R9 = (12V - 1.2V) / 4.7kOhm = 2.3mA)
R8 = 10kOhm LV output resistor (IC_sat = 12V / R8 = 12V / 10kOhm = 1.2mA <= IQ7_LED * CTRmin)
C10, C11, C12 = 0.1uF as datasheet directions (multilayer ceramic)
Hardware-only precharge controller design for the E-Agle Trento Racing Team electronics recruitment test.
By Bertagnolli Marco
University of Trento
Sheet: /hv_sense/
File: hv_sense.kicad_sch
Title: Hardware precharge-controller
Size: A4
Date:
Rev: 1.0
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Input: V_{TS} = 0V to 600V DC

Max current draw (HV precharge branch):
$I_{pre_peak} = V_{TS_max} / R17 = 600V / 10000Ohm = 0.60A$ (< 1A)

Sizing HV Relays & Precharge Resistor:
U2, U3, U4 = Omron G9KB-1A12V ($V_{contact_max}$ = 600V DC, V_{coil} = 12V DC, R_{coil} = 510hm)
R_{pre} = 10000hm Precharge Power Resistor
tau = R17 * C_{DC_LINK} = 10000hm * 630uF = 0.63s (or 0.378s with C_{DC_LINK} = 378uF)
$t_{95\%}$ = tau * ln(20) = 3 * tau = 1.89s (or 1.13s with C_{DC_LINK} = 378uF)

Hardware-only precharge controller design for the E-Agle Trento Racing Team electronics recruitment test.

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Sheet: /hv_power/
File: hv_power.kicad_sch

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Size: A4	Date:	**Rev: 1.0**
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